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SWITCH REGULATOR

Abstract

A circuit includes an amplifier, a first transistor, a second transistor, and a third transistor. The amplifier has a first amplifier input coupled to a voltage reference terminal; a second amplifier input, and an amplifier output. The first transistor has a first terminal coupled to a power terminal, a second terminal coupled to the second amplifier input, a body terminal coupled to a reference terminal, and a control terminal coupled to the amplifier output. The second transistor has a first terminal coupled to the power terminal, a second terminal, a body terminal coupled to the second terminal of the second transistor, and a control terminal coupled to the amplifier output. The third transistor has a first terminal coupled to the second amplifier input, a second terminal coupled to the reference terminal, and a control terminal coupled to the second terminal of the second transistor.

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Background/Summary

BACKGROUND

[0001] Digital-to-analog converters (DACs) are used in a variety of applications to convert a digital value to an analog signal. For example, a DAC may be used with, or integrated into, a microcontroller or microprocessor to convert digital values to analog signals. Some DACs include a resistor ladder having multiple resistors, e.g., SiCr thin-film or poly-silicon resistors, to generate an analog signal representing a digital value.

SUMMARY

[0002] In one example, a circuit includes an amplifier, a first transistor, a second transistor, and a third transistor. The amplifier has a first amplifier input coupled to a voltage reference terminal, a second amplifier input, and an amplifier output. The first transistor having a first terminal coupled to a power terminal, a second terminal coupled to the second amplifier input, a body terminal coupled to a reference terminal, and a control terminal coupled to the amplifier output. The second transistor has a first terminal coupled to the power terminal, a second terminal, a body terminal coupled to the second terminal of the second transistor, and a control terminal coupled to the amplifier output. The third transistor has a first terminal coupled to the second amplifier input, a second terminal coupled to the reference terminal, and a control terminal coupled to the second terminal of the second transistor.

[0003] In another example, a circuit includes an amplifier, a resistor, and first, second, third, fourth, fifth, and sixth transistors. The amplifier has a first amplifier input coupled to a voltage reference terminal; a second amplifier input, and an amplifier output. The first transistor has a first terminal coupled to a power terminal; a second terminal coupled to the second amplifier input, a body terminal coupled to a reference terminal, and a control terminal coupled to the amplifier output. The second transistor has a first terminal coupled to the power terminal; a second terminal, a body terminal coupled to the second terminal of the second transistor, and a control terminal coupled to the amplifier output. The third transistor has a first terminal coupled to the second amplifier input, a second terminal, and a control terminal coupled to the second terminal of the second transistor. The resistor has a first terminal coupled to the second terminal of the second transistor, and a second terminal. The fourth transistor has a first terminal, a second terminal, and a control terminal coupled to the second terminal of the second transistor. The fifth transistor has a first terminal, a second terminal, and a control terminal coupled to the first terminal of the fourth transistor. The sixth transistor has a first terminal coupled to the second terminal of the fifth transistor, a second terminal coupled to the reference terminal, and a control coupled to the second terminal of the resistor.

[0004] In a further example, a digital-to-analog converter includes a resistor ladder, a switch circuit, a first driver circuit, a second driver circuit, and a switch regulator circuit. The switch circuit is coupled to a rung of the resistor ladder. The switch circuit includes a first switch having a control input and a second switch having a control input. The first driver circuit has an output coupled to the control input of the first switch. The second driver circuit has an output coupled to the control input of the second switch, and a voltage input. The switch regulator circuit has an output coupled to the voltage input of the second driver circuit. The switch regulator circuit includes an amplifier, a first transistor, a second transistor, and a third transistor. The amplifier has a first amplifier input coupled to a voltage reference terminal, a second amplifier input, and an amplifier output. The first transistor has a first terminal coupled to a power terminal, a second terminal coupled to the second amplifier input, a body terminal coupled to a reference terminal, and a control terminal coupled to the amplifier output. The second transistor has a first terminal coupled to the power terminal, a second terminal, a body terminal coupled to the second terminal of the

second transistor, and a control terminal coupled to the amplifier output. The third transistor has a first terminal coupled to the second amplifier input, a second terminal coupled to the reference terminal, and a control terminal coupled to the second terminal of the second transistor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a schematic diagram of an example digital-to-analog converter (DAC).

[0006] FIG. 2 is a schematic diagram of a switch circuit suitable for use in the DAC of FIG. 1.

[0007] FIG. 3 is a schematic diagram of a switch regulator circuit suitable for use with the switch circuit of FIG. 2.

[0008] FIG. 4 is a block diagram of an example system that includes the switch regulator circuit of FIG. 3.

DETAILED DESCRIPTION

[0009] FIG. 1 is a schematic diagram of an example digital-to-analog converter (DAC) **100**. The DAC **100** includes a resistor ladder **101**, switches **105**, and an amplifier **108**. The resistor ladder **101** includes resistors **104A**, **104B**, **104C**, **104D**, and **104E** forming the rungs of the ladder, and resistors **102A**, **102B**, and **102C** forming the rail of the ladder. The resistors **102A**, **102B**, and **102C** may each have resistance R , and the resistors **104A**, **104B**, **104C**, **104D**, and **104E** may each have resistance $2R$.

[0010] The resistor ladder **101** is coupled to the switches **105** and the amplifier **108**. The resistor ladder **101** binary weights and sums the voltages provided by the switches **105** to generate an output voltage that is provided to the amplifier **108**. The amplifier **108** is configured as a voltage follower with a non-inverting input coupled to the resistor ladder **101** and an inverting input coupled to the output of the amplifier **108**. The output of the amplifier **108** may be coupled to any circuitry that uses the output signal ($V_{\text{sub.OUT}}$) generated by the DAC **100**.

[0011] The switches **105** includes switches **106A**, **106B**, **106C**, and **106D** that are respectively coupled to resistors **104B**, **104C**, **104D**, and **104E**. The switches **106A**, **106B**, **106C**, and **106D** are illustrated as single-pole double-throw switches having a first terminal coupled to a resistor **104** (e.g., one of the resistors **104A**, **104B**, **104C**, or **104D**), a second terminal coupled to a voltage reference terminal (V_{REF}), and a third terminal coupled to ground. The switch **106D** represents the most significant bit of digital value and the switch **106A** represents the least significant bit of the digital value. Each switch **106** (e.g., switch **106A**, **106B**, **106C**, or **106D**) connects the first terminal to the second terminal if the bit of the digital value represented by the switch is a logic one, and connects the first terminal to the third terminal if the bit is a logic zero.

[0012] While the DAC **100** is illustrated as a 4-bit DAC, examples of the DAC **100** may include any number of the resistors **102**, resistors **104**, and switches **106** to form a DAC having a different number of bits (e.g., an 8-bit DAC, 10-bit DAC, 12-bit DAC, 16-bit DAC, etc.).

[0013] FIG. 2 is a schematic diagram of a switch **106** suitable for use in the DAC **100**. As explained above, the switch **106** is a single-pole double-throw switch. The switch **106** includes transistors **202** and **204** to provide single-pole double-throw operation. The transistor **202** and the transistor **204** may be n-type field effect transistors (NFETs). A first terminal (e.g., drain) of the transistor **202** is coupled to a resistor **104** of the resistor ladder **101**, and to a first terminal (e.g., drain) of the transistor **204**. A second terminal (e.g., source) of the transistor **202** is coupled to ground (or other reference voltage terminal representing a logic zero (V_{REFL})), and a second terminal of the transistor **204** is coupled to a voltage reference terminal that provides a voltage (V_{REFH}) representing a logic one. A control terminal (e.g., gate) of the transistor **202** is coupled to a driver circuit **206** that provides a control signal (V_{SWL}) for turning the transistor **202** on or off. For example, V_{SWL} turns the transistor **202** on if the switch **106** is to provide a logic zero to the

resistor **104**, and turns the transistor **202** off if a logic one is to be provided to the resistor **104**. A control terminal (e.g., gate) of the transistor **204** is coupled to a driver circuit **208** that provides a control signal (VSWH) for turning the transistor **204** on or off. For example, VSWH turns the transistor **204** on if the switch **106** is to provide a logic one to the resistor **104**, and turns the transistor **204** off if a logic zero is to be provided to the resistor **104**.

[0014] In high resolution and high accuracy DACs (e.g., 16-bit resolution and above), the impedance of the switch **106** should be very low and the resistance of the transistor **202** and transistor **204** should be well matched. Accordingly, the transistor **202** and the transistor **204** may have a relatively large W/L size, and be relatively equal in size. The VSWL provided to the transistor **202** may swing to a power supply voltage (AVDD) as the source of the transistor **202** is at ground. However, VSWH provided to the transistor **204** may swing to AVDD-VREF as when on, its source voltage will be at VREF. Accordingly, in some examples of the switches **105**, the VSWL may also swing to AVDD-VREF to match VSWH, and with the transistor **202** and transistor **204** having the same size and control voltage, the ideal impedance of the transistor **202** and transistor **204** will be the same, barring any process mismatch in the transistors.

[0015] However, the impedances of the transistor **202** and transistor **204** will not be the same with equal gate-to-source (VGS) control voltage if one of the transistors is subject to a back-body voltage effect. The back-body voltage effect, which may also be referred to as body voltage effect or back-gate effect, is a phenomenon whereby the voltage applied to a body terminal of a transistor affects the threshold voltage of the transistor. The back-body voltage effect can be reduced or eliminated simply by tying the well or body terminal of the transistor to its source. This is easily done for a ground potential switch as all GND potential switches are in a common PWELL. The transistor **202** has a back-body terminal that is coupled to the source of the transistor **202**. However, for VREF referenced switches such as the transistor **204**, to tie the PWELL body terminal to its source requires that the VREF connected switch be isolated (e.g., the body of the transistor be in its own isolated PWELL). Such isolation in some fabrication processes requires a significant increase in the area of the transistor and is therefore impractical. In the switch **106**, the body terminal of the transistor **204** is coupled to ground, similar to the transistor **202**. While this significantly reduces the layout area of the transistor **204**, it does however introduce a VBS or body voltage effect related error in the impedance of the transistor **204**. The body-to-source voltage for the transistor **204** can be as much as VREF (e.g., 2.5 volts), creating an impedance mismatch that would reduce accuracy to less than 16-bit performance levels.

[0016] FIG. **3** is a schematic diagram of a switch regulator circuit **300** suitable for use with the switch **106**. The switch regulator circuit **300** overcomes the body voltage effect of the transistor **204** in regard to its impedance match with the transistor **202**. The switch regulator circuit **300** determines the actual body voltage effect and adds it as a correction to the control voltage provided to the transistor **202**. While the impedance of the transistor **204** will be higher due to the body voltage effect thereof, the gate drive of the transistor **202** is reduced from its nominal AVDD-VREF voltage to an AVDD-VREF- Δ VBS effect voltage to match impedances of the transistor **202** and the transistor **204**.

[0017] The switch regulator circuit **300** includes a regulator core circuit **302** and an output circuit **304** coupled to the regulator core circuit **302**. The regulator core circuit **302** generates an output that includes the back-body voltage effect of the transistor **204**. The output circuit **304** generates a voltage output that is a power supply voltage (AVDD) less the voltage received from the regulator core circuit **302**. The output voltage of the output circuit **304** can be used to control the transistor **202**, such that the impedance of the transistor **202** is increased to match the impedance of the transistor **204**.

[0018] The regulator core circuit **302** includes an amplifier **306**, transistors **308**, **310**, **312**, **316**, **318**, and **322**, resistors **314** and **320**, and a voltage source **336** (which provides a small voltage bias). A first amplifier input (e.g., a non-inverting input) of the amplifier **306** is coupled to a voltage

reference terminal for receipt of VREF. An amplifier output of the amplifier **306** is coupled to a control terminal (e.g., gate) of the transistor **308**. A body terminal of the transistor **308** is coupled to a reference voltage terminal (e.g., ground). A first current terminal (e.g., drain) of the transistor **308** is coupled to a power terminal providing AVDD via the transistor **316**. A second terminal (e.g., source) of the transistor **308** is coupled to a second amplifier input (e.g., an inverting input) of the amplifier **306**.

[0019] The output of the amplifier **306** is also coupled to a control terminal (e.g., gate) of the transistor **310**. A first current terminal (e.g., drain) of the transistor **310** is coupled to the power terminal providing AVDD via the transistor **318**. A second terminal (e.g., source) of the transistor **310** is coupled to a body terminal of the transistor **310**.

[0020] The transistor **312** provides feedback from the transistor **310** to the amplifier **306**. A control terminal (e.g., gate) of the transistor **312** is coupled to the second terminal of the transistor **310**. A first terminal (e.g., drain) of the transistor **312** is coupled to the second input of the amplifier **306** and the second terminal of the transistor **308**. A second terminal of the transistor **312** is coupled to a reference voltage terminal (e.g., AVSS, ground) via the resistor **314**. The resistor **314** has a first terminal coupled to the second terminal of the transistor **312** and a second terminal coupled to the reference voltage terminal.

[0021] The resistor **320** and the transistor **322** are coupled in series between the second terminal of the transistor **310** and the reference voltage terminal. The transistor **322** is diode connected with a control terminal (e.g., gate) of the transistor **322** coupled to a first terminal (e.g., drain) of the transistor **322**. A second terminal (e.g., source) of the transistor **322** is coupled to the reference voltage terminal. The resistor **320** has a first terminal coupled to the second terminal of the transistor **310** and a second terminal coupled to the first terminal of the transistor **322**. The first and second terminals of the resistor **320** may also be coupled to other circuits, e.g., other instances of the output circuit **304** as shown in FIG. 4.

[0022] The transistor **316** is coupled between the power terminal and the transistor **308**. A first terminal (e.g., drain) of the transistor **316** is coupled to the power terminal. A second terminal (e.g., source) of the transistor **316** is coupled to the first terminal of the transistor **308**. A control terminal (e.g., gate) of the transistor **316** is coupled to the output of the amplifier **306** via the voltage source **336**. The voltage source **336** may include one or more diode-connect transistors or a resistor and a current source coupled to the power terminal. The transistor **318** is coupled between the power terminal and the transistor **310**. A first terminal of the transistor **318** is coupled to the power terminal. A second terminal of the transistor **318** is coupled to the first terminal of the transistor **310**. A control terminal of the transistor **318** is coupled to the control terminal of the transistor **316**.

[0023] The output circuit **304** includes transistors **326**, **332** and **334**, resistors **324** and **328**, and a capacitor **330**. A control terminal (e.g., gate) of the transistor **326** is coupled to the second terminal of the transistor **310**. A first terminal (e.g., drain) of the transistor **326** is coupled to the power terminal via the resistor **324**. A second terminal (e.g., source) of the transistor **326** is coupled to the reference voltage terminal via the resistor **328**. The resistor **324** has a first terminal coupled to the power terminal and a second terminal coupled to the first terminal of the transistor **326**. The resistor **328** has a first terminal coupled to the second terminal of the transistor **326** and a second terminal coupled to the reference voltage terminal.

[0024] The transistor **332** includes a first terminal (e.g., drain) coupled to the power terminal, and a control terminal (e.g., gate) coupled to the first terminal of the transistor **326**. A second terminal (e.g., source) of the transistor **332** is coupled to a first terminal (e.g., drain) of the transistor **334**. The reference voltage output (VREGA) of the switch regulator circuit **300** is provided at the second terminal of the transistor **332**. The second terminal of the transistor **332** may be coupled to a voltage input of the driver circuit **206** (FIG. 2) so that the driver circuit **206** applies VREGA to control the transistor **202**. A second terminal (e.g., source) of the transistor **334** is coupled to the reference voltage terminal, and a control terminal (e.g., gate) of the transistor **334** is coupled to the

first terminal of the transistor **322**. The capacitor **330** has a first terminal coupled to the control terminal of the transistor **332** and a second terminal coupled to the reference voltage terminal. [0025] In operation, the amplifier **306** forces the second terminal of the transistor **308** to be at a same voltage as the first terminal of the amplifier **306** (e.g., VREF). The transistor **308** and the transistor **310** have a same control voltage, the output voltage of the amplifier **306**. However, transistor **308**, having its body terminal at GND will experience an increased VGS relative to transistor **310**, which does not have any body voltage effect as the body terminal of the transistor **310** is connected back to its source. Since the gate voltage of transistors **308** and **310** is the same, the voltage seen at the source of transistor **310** will be higher than the voltage at the source of the transistor **308** by the difference in the body voltage effect of the transistors **308** and **310**. This can be seen by analyzing the operation of the switch regulator circuit **300** starting at the source voltage of transistor **308**. Transistor **308** source voltage will be held at VREF by the amplifier **306**. The output voltage of the amplifier **306** and the gate voltage of the transistor **308** will therefore be VREF+VGS.sub.308 (the VGS of the transistor **308**). But since VGS.sub.308 is essentially equal to the VGS.sub.310 but with the increase of the body voltage effect on the transistor **308**, VGS.sub.308 can be written as:

$$[00001] \text{VGS}_{308} = \text{VGS} + \text{VBSeffect} \quad (1)$$

where VGS is the gate-to-source voltage of the transistor **310**.

[0026] The voltage across the resistor **320** is:

$$[00002] \text{VREF} + \text{VGS}_{308} - \text{VGS}_{310} - \text{VGS}_{322} \quad (2)$$

[0027] The VGSs of the transistor **310** and the transistor **322** are the same, and the VGS of the transistor **308** is as per equation (1), allowing reduction of equation (2) to:

$$[00003] \text{VREF} + \text{VBSeffect} - \text{VGS} \quad (3)$$

[0028] The voltage across the resistor **314** is the same as the voltage across the resistor **320**, and the current flowing through the resistor **314** is the same as the current flowing through the resistor **320**. Accordingly, currents flowing through the transistor **308** and the transistor **310** are equal. The current flowing through the resistor **320** is:

$$[00004] \frac{\text{VREF} + \text{VBSeffect} - \text{VGS}}{\text{R2}} \quad (4)$$

where R2 is the resistance of the resistor **320**.

[0029] The current flowing through the resistor **328** is the same as the current flowing through the resistor **320** (equation (4)), and the resistance of the resistor **320**, the resistor **324**, and the resistor **328** may be the same. Accordingly, the voltage at the first terminal of the transistor **326** and the gate of the transistor **332** may be expressed as:

$$[00005] \text{AVDD} - \text{R3} \frac{\text{VREF} + \text{VBSeffect} - \text{VGS}}{\text{R2}} \quad (5)$$

where AVDD is the voltage provided at the power supply terminal, and R3 is the resistance of the resistor **324**.

[0030] Note that the voltage of equation (5) contains the AVDD-VREF relation, but also includes the ΔVBSeffect voltage. Finally, the output voltage of the switch regulator circuit **300** provided at the source of the transistor **322** is:

$$\text{AVDD} - \text{VREF} - \Delta \text{VBSeffect} \quad (6)$$

providing a proper regulator voltage for the transistor **202** to match the impedance of the transistor **204**.

[0031] FIG. 4 is a block diagram of an example system **400** that includes the switch regulator circuit **300**. The system **400** includes the regulator core circuit **302**, multiple instances of the output circuit **304**, multiple driver circuits **402**, and multiple DACs **404**. The output of the regulator core circuit **302** (e.g., taken at the first and second terminals of the resistor **320** shown in FIG. 3) is coupled to each output circuit **304** (e.g., coupled to the control terminals of the transistor **326** and

344 as shown in FIG. 3). Each instance of the output circuit 304 is coupled to a voltage input of an instance of the driver circuit 402, and an output of each instance of the driver circuit 402 is coupled to an instance of the DAC 404. The driver circuit 402 may be an implementation of the driver circuit 206. The driver circuit 402 provides the control signal VSWL, which swings to VREG provided by the output circuit 304, for controlling the switch 106 in the DAC 404. In the system 400, a single regulator core circuit 302 provides a voltage output for controlling any number of instances of the DAC 404, which reduces the circuitry of the system 400 and provides improved linearity in the DACs 404.

[0032] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0033] As used herein, the terms “terminal,” “node,” “interconnection,” “pin” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component.

[0034] A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

[0035] While the use of particular transistors is described herein, other transistors (or equivalent devices) may be used instead with little or no change to the remaining circuitry. For example, a field effect transistor (“FET”) (such as an n-channel FET (NFET) (n-type transistor) or a p-channel FET (PFET)) (p-type transistor)), a bipolar junction transistor (BJT—e.g., NPN transistor or PNP transistor), an insulated gate bipolar transistor (IGBT), and/or a junction field effect transistor (JFET) may be used in place of or in conjunction with the devices described herein. The transistors may be depletion mode devices, drain-extended devices, enhancement mode devices, natural transistors, or other types of device structure transistors. Furthermore, the devices may be implemented in/over a silicon substrate (Si), a silicon carbide substrate (SiC), a gallium nitride substrate (GaN) or a gallium arsenide substrate (GaAs).

[0036] References may be made in the claims to a transistor's control input and its current terminals. In the context of a FET, the control input (or transistor control terminal) is the gate, and the current terminals are the drain and source. In the context of a BJT, the control input is the base, and the current terminals are the collector and emitter.

[0037] References herein to a FET being “ON” means that the conduction channel of the FET is present and drain current may flow through the FET. References herein to a FET being “OFF” means that the conduction channel is not present so drain current does not flow through the FET. An “OFF” FET, however, may have current flowing through the transistor's body-diode.

[0038] Circuits described herein are reconfigurable to include additional or different components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance

represented by the resistor shown. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor.

[0039] While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other example embodiments, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated. As used herein, the term “integrated circuit” means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.

[0040] Uses of the phrase “ground” in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within ± 10 percent of that parameter or, if the parameter is zero, a reasonable range of values around zero.

[0041] Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. A circuit comprising: an amplifier having a first amplifier input coupled to a voltage reference terminal; a second amplifier input, and an amplifier output; a first transistor having a first terminal coupled to a power terminal; a second terminal coupled to the second amplifier input; a body terminal coupled to a reference terminal; and a control terminal coupled to the amplifier output; a second transistor having a first terminal coupled to the power terminal; a second terminal; a body terminal coupled to the second terminal of the second transistor; and a control terminal coupled to the amplifier output; and a third transistor having a first terminal coupled to the second amplifier input; a second terminal coupled to the reference terminal; and a control terminal coupled to the second terminal of the second transistor.
2. The circuit of claim 1, further comprising a first resistor having a first terminal coupled to the second terminal of the third transistor and a second terminal coupled to the reference terminal.
3. The circuit of claim 1, further comprising: a fourth transistor having a first terminal, a second terminal coupled to the reference terminal, and a control terminal coupled to the first terminal of the fourth transistor; and a first resistor having a first terminal coupled to the second terminal of the second transistor and a second terminal coupled to the first terminal of the fourth transistor.
4. The circuit of claim 1, further comprising: a fourth transistor having a first terminal; a second terminal, and a control terminal coupled to the second terminal of the second transistor; a first resistor having a first terminal coupled to the power terminal and a second terminal coupled to the first terminal of the fourth transistor; and a second resistor having a first terminal coupled to the second terminal of the fourth transistor and a second terminal coupled to the reference terminal.
5. The circuit of claim 4, further comprising: a fifth transistor having a first terminal coupled to the power terminal, a second terminal, and a control terminal coupled to the first terminal of the fourth transistor; and a sixth transistor having a first terminal coupled to the second terminal of the fifth transistor, a second terminal coupled to the reference terminal, and a control terminal coupled to the second terminal of the second transistor.

6. The circuit of claim 4, further comprising a capacitor having a first terminal coupled to the first terminal of the fourth transistor and a second terminal coupled to the reference terminal.
7. The circuit of claim 1, further comprising: a fourth transistor having a first terminal coupled to the power terminal, a second terminal coupled to the first terminal of the first transistor, and a control terminal; a fifth transistor having a first terminal coupled to the power terminal, a second terminal coupled to the first terminal of the second transistor, and a control terminal coupled to the control terminal of the fourth transistor; and a voltage source having a first terminal coupled to the amplifier output and a second terminal coupled to the control terminal of the fifth transistor.
8. A circuit comprising: an amplifier having a first amplifier input coupled to a voltage reference terminal; a second amplifier input, and an amplifier output; a first transistor having a first terminal coupled to a power terminal; a second terminal coupled to the second amplifier input, a body terminal coupled to a reference terminal, and a control terminal coupled to the amplifier output; a second transistor having a first terminal coupled to the power terminal; a second terminal, a body terminal coupled to the second terminal of the second transistor and a control terminal coupled to the amplifier output; a third transistor having a first terminal coupled to the second amplifier input; a second terminal, and a control terminal coupled to the second terminal of the second transistor; a resistor having a first terminal coupled to the second terminal of the second transistor, and a second terminal; a fourth transistor having a first terminal, a second terminal, and a control terminal coupled to the second terminal of the second transistor; a fifth transistor having a first terminal, a second terminal, and a control terminal coupled to the first terminal of the fourth transistor; and a sixth transistor having a first terminal coupled to the second terminal of the fifth transistor, a second terminal, and a control coupled to the second terminal of the resistor.
9. The circuit of claim 8, wherein: the resistor is a first resistor; and the circuit includes: a seventh transistor having a first terminal coupled to the second terminal of the first resistor, a second terminal coupled to the second terminal of the sixth transistor, and a control terminal coupled to the first terminal of the seventh transistor; and a second resistor having a first terminal coupled to the second terminal of the third transistor and a second terminal coupled to the second terminal of the seventh transistor.
10. The circuit of claim 8, further comprising a seventh transistor having a first terminal coupled to a power terminal, a second terminal coupled to the first terminal of the second transistor, and a control terminal coupled to the amplifier output.
11. The circuit of claim 10, further comprising an eighth transistor having a first terminal coupled to the power terminal, a second terminal coupled to the first terminal of the first transistor, and a control terminal coupled to the control terminal of the seventh transistor.
12. The circuit of claim 10, further comprising a voltage source having a first terminal coupled to the amplifier output and a second terminal coupled to the control terminal of the seventh transistor.
13. The circuit of claim 8, wherein: the resistor is a first resistor; and the circuit includes: a second resistor having a first terminal coupled to the first terminal of the fifth transistor and a second terminal coupled to the first terminal of the fourth transistor; and a third resistor having a first terminal coupled to the second terminal of the fourth transistor and a second terminal coupled to the second terminal of the sixth transistor.
14. The circuit of claim 8, further comprising a capacitor having a first terminal coupled to the control terminal of the fifth transistor and a second terminal coupled to the second terminal of the sixth transistor.
15. The circuit of claim 8, wherein: the resistor is a first resistor; and the circuit includes a second resistor having a first terminal coupled to the second terminal of the third transistor and a second terminal coupled to the second terminal of the sixth transistor.
16. A digital-to-analog converter comprising: a resistor ladder; a switch circuit coupled to a rung of the resistor ladder, the switch circuit including a first switch having a control input and a second switch having a control input; a first driver circuit having an output coupled to the control input of

the first switch; a second driver circuit having an output coupled to the control input of the second switch, and a voltage input; a switch regulator circuit having an output coupled to the voltage input of the second driver circuit, the switch regulator circuit including: an amplifier having a first amplifier input coupled to a voltage reference terminal; a second amplifier input, and an amplifier output; a first transistor having a first terminal coupled to a power terminal; a second terminal coupled to the second amplifier input, a body terminal coupled to a reference terminal, and a control terminal coupled to the amplifier output; a second transistor having a first terminal coupled to the power terminal; a second terminal, a body terminal coupled to the second terminal of the second transistor, and a control terminal coupled to the amplifier output; and a third transistor having a first terminal coupled to the second amplifier input, a second terminal coupled to the reference terminal, and a control terminal coupled to the second terminal of the second transistor.

17. The digital-to-analog converter of claim 16, further comprising a first resistor having a first terminal coupled to the second terminal of the third transistor and a second terminal coupled to the reference terminal.

18. The digital-to-analog converter of claim 16, further comprising: a fourth transistor having a first terminal, a second terminal coupled to the reference terminal, and a control terminal coupled to the first terminal of the fourth transistor; and a first resistor having a first terminal coupled to the second terminal of the second transistor and a second terminal coupled to the first terminal of the fourth transistor.

19. The digital-to-analog converter of claim 16, further comprising: a fourth transistor having a first terminal; a second terminal, and a control terminal coupled to the second terminal of the second transistor; a second resistor having a first terminal coupled to the power terminal and a second terminal coupled to the first terminal of the fourth transistor; and a third resistor having a first terminal coupled to the second terminal of the fourth transistor and a second terminal coupled to the reference terminal.

20. The digital-to-analog converter of claim 19, further comprising: a fifth transistor having a first terminal coupled to the power terminal, a second terminal coupled to the voltage input of the second driver circuit, and a control terminal coupled to the first terminal of the fourth transistor; and a sixth transistor having a first terminal coupled to the second terminal of the fifth transistor, a second terminal coupled to the reference terminal, and a control terminal coupled to the second terminal of the second transistor.
